

F24-100-D-Palantir360

Project Team

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Chapter 1

Introduction

Palantir360 is a development-based project that aims to make real estate buying easier for people who cannot visit on-site. Palantir360 bridges the gap between buyers and their potential investments by offering immersive virtual tours and detailed property insights no matter where they are located. It eliminates the need for time-consuming visits, helping users make informed decisions with confidence. The platform is designed to cater to both local and international buyers, providing an accessible experience for all. With Palantir360, real estate transactions are more user-friendly than ever before. [?].

1.1 Problem Statement

Imagine you are sitting overseas, and you want to buy a property in a society in Pakistan. But you just have images and videos of the property at your disposal. The details are vague or sometimes incomplete and you can't fully experience the dynamics of the property. Now with Palantir360 you can visit the property virtually from the comfort of your own home.

1.2 Scope

Our Palantir360 will focus on the creation of an immersive experience for both indoor and outdoor navigation of houses in real estate societies. Our core objective is to develop a detailed 3D visualisation of the interior of homes, allowing users to explore the space in VR through character movement. This navigation will provide an in-depth view of both the layout and the design of the property. For the end users, this would mean an engaging and realistic experience of moving through different rooms and spaces within the house. [?]

But this is not all, Palantir360 will also blend outdoor impressions visible from the windows, giving users a glimpse of the house's external surroundings without needing to cover expansive outdoor areas. This would help to provide context to the buyer about the house's location and setting while keeping the emphasis on immediate environment visible from inside. The combination of detailed inside exploration with the subtle outdoor views will help to create a balanced experience for users interested in remote property viewing.

1.3 Modules

Write down the modules of the proposed project. Each module should highlight features, using bulleted/numbered notation. When developing both a mobile app and a web app, group the modules according to the system types, such as, Client Web App, Client Mobile App, Admin Web App etc.

1.3.1 VR UI Module

Implementing action-based object interaction. For example, selecting items, object rotation as well as zoom in and out. Adjustment of the UI with respect to the to object proximity from the users, and to integrate optimal visibility from their line of sight.

1. Adaptive UI
2. Motion-Based Interaction

1.3.2 3D Environment Modelling

Creating indoor 3D room models and the outdoor environment adding dynamic weather like cloudy or sunny conditions to enhance environment's realism. A realistic day and night cycle along with light and sound effects.

1. Room Model Creation
2. Day/Night Shift and Weather System

1.3.3 VR Navigation Module

Facilitate customersto navigate real-estate terrain like walking on floor or stairs for a more realistic walkthrough feel. Implementing the teleportation within the environment and avoid teleportation in unwanted mesh.

1. Location-based Teleportation
2. Human-like Navigation

1.3.4 Mini-map Integration

Display and highlight the information in the VR environment when the user clicks on the desired region on the map. Facilitate the user by providing the guided paths on the map to walk through the properties.

1. Interactive Landmarks
2. Wayfinding Assistance

1.3.5 3D Floor Structuring Module

Create detailed and interactive 3D floor plans before integrating room scans to the building structure. Use architectural data to create accurate 3D representation of the layout of a property.

1. Positioning and Scaling of Room Scans
2. Real-time Measurements

1.3.6 VR Optimization and Performance Module

Adjusting the loading times based on the level of details in the VR environment. While the user is navigating through the property, the assets will be loaded in the background to minimize delay in loading.

1. Adaptive Loading
2. Asynchronous Loading

1.4 User Classes and Characteristics

User class	Description
Potential Home Buyers and Renters	Individuals seeking properties for personal or investment purposes. These users will use Palantir360 to explore and navigate through properties via 3D and VR tours, allowing them to make more informed decisions without being physically present.
Real Estate Agents	Professionals who showcase properties to clients and assist in the buying process. These users rely on Palantir360 to present properties more effectively through immersive VR tours and navigation, helping clients visualize and assess potential investments.
Real Estate Developers	Developers interested in promoting new housing developments and showcasing properties through VR tours. These users use Palantir360 to create engaging and realistic virtual representations of their developments, enabling potential buyers to explore properties remotely.
Virtual Tour Providers (Tech Partners)	Technical teams responsible for integrating VR tours into their real estate platforms (Palantir360). These users focus on ensuring seamless integration and functionality of VR tours, as well as providing support and technical expertise.

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case Diagram

We have chosen Use Case Diagram along with High Level Use Cases as they are the most suitable representation for our project.

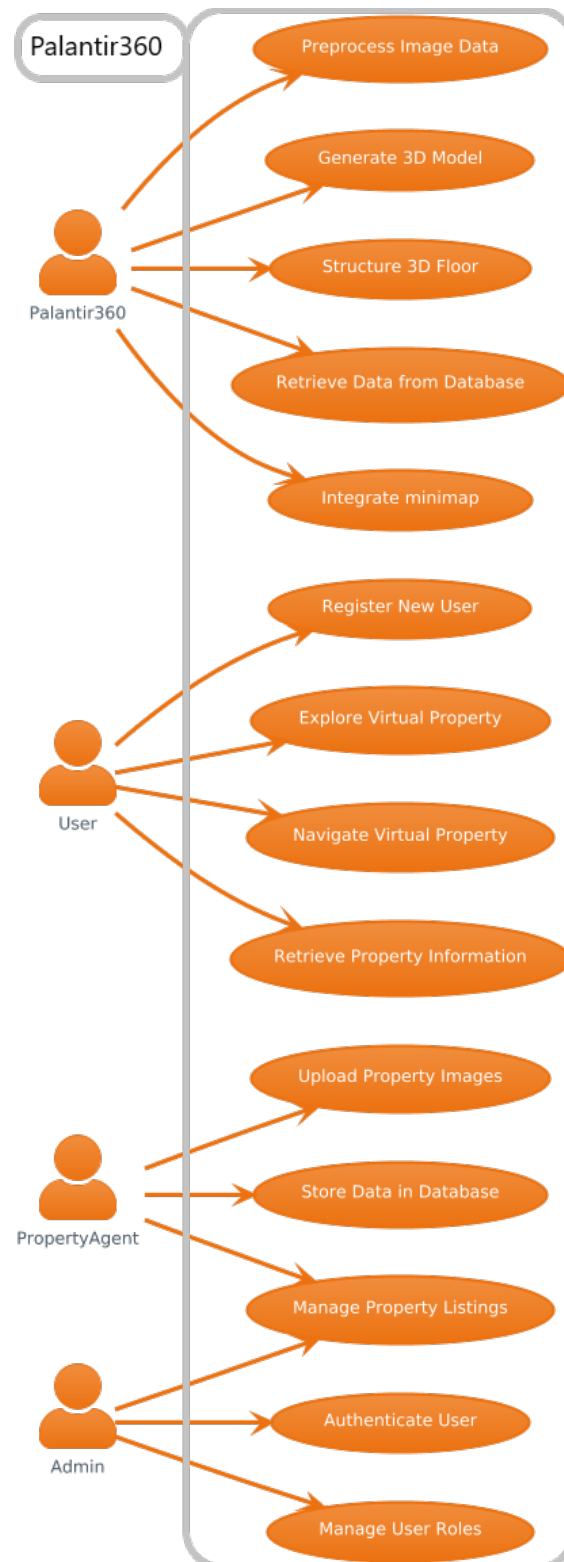


Figure 2.1: Use-Case Diagram.

2.2 High-Level Use Cases

Here are the high-level use cases for the Palantir360 project based on the analysis of its documentation and workflow:

2.2.1 User Registration and Authentication

- **Actors:** Home buyers, renters, real estate agents, developers
- **Precondition:** The user must have access to the mobile app or VR system.
- **Description:** Users can create an account by providing personal details, such as name, email, and password. After registration, users can log in to access the platform.
- **Main Success Scenario:**
 - The user provides valid registration details.
 - The system verifies and stores user information.
 - The user successfully logs in.
- **Alternate Scenarios:**
 - Registration with invalid information results in a failure message.
 - Failed login attempts after multiple tries prompt the user to reset their password.
- **Postcondition:** Users gain access to the system's features (e.g., virtual property tours).

2.2.2 Upload Property Images (Agent Module)

- **Actors:** Real estate agents, developers
- **Precondition:** The user must be logged in and have property images ready for upload.
- **Description:** Real estate agents can upload images of properties to the database, which will be processed for 3D model generation.
- **Main Success Scenario:**
 - The user selects and uploads high-quality images.
 - The system validates and processes images for compatibility (resolution, focus, lighting).
 - Images are stored in the Firebase database for further use.

- **Alternate Scenarios:**

- Images fail validation (blurry, too dark) and are rejected for re-upload.

- **Postcondition:** Property images are successfully uploaded and stored.

2.2.3 Image Data Preprocessing

- **Actors:** System

- **Precondition:** Uploaded images must be available in the database.

- **Description:** The system processes the images, optimizing them for the 3D model generation. The system identifies and corrects issues like blurriness, improper lighting, and edge detection.

- **Main Success Scenario:**

- The system analyzes uploaded images.
- The system enhances image quality or flags problematic images for re-upload.
- Optimized images are stored for model generation.

- **Postcondition:** Image data is optimized and ready for 3D model generation.

2.2.4 3D Model Generation

Actors: System (AutoDesk API/ NERF)

Precondition: Image data must be preprocessed and available in the database.

Description: The system retrieves processed images and uses Autodesk's API / NERF API to generate 3D models of the property.

Main Success Scenario:

- The system retrieves images from the database.
- The AutoDesk / NERF API [2] processes the images and generates a 3D model.
- The 3D model is stored in the database.

Alternate Scenarios:

- Failed API call due to incomplete or corrupt data.

Postcondition: A 3D model of the property is generated and stored.

2.2.5 Virtual Property Exploration (VR UI Module)

Actors: Home buyers, renters

Precondition: A 3D model must be available in the database.

Description: Users can view and interact with virtual property models through the VR environment, exploring both indoor and outdoor spaces.

Main Success Scenario:

- The user navigates through the property using VR controllers.
- Interactive features (zoom, object rotation, view adjustment) enhance the experience.

Alternate Scenarios:

- Users experience lags or incomplete model loading during navigation.

Postcondition: Users successfully navigate through the virtual property in the VR system.

2.2.6 Navigation through Property (VR Navigation Module)

Actors: Home buyers, renters

Precondition: A 3D model of the property must be instantiated in the VR environment.

Description: Users navigate the virtual space through human-like movement, including walking up stairs and teleporting to different locations.

Main Success Scenario:

- The user moves naturally through the space with realistic physics.
- Teleportation is enabled for distant locations.

Alternate Scenarios:

- Navigation fails or users experience unnatural movement.

Postcondition: The user explores the property with smooth, human-like movement.

2.2.7 Mini-Map and Wayfinding Integration

Actors: Home buyers, renters

Precondition: The user must be inside a virtual property.

Description: A mini-map shows the layout of the property and assists the user in navigating the space. Interactive landmarks allow for quick teleportation.

Main Success Scenario:

- The user opens the mini-map.
- The user selects landmarks or paths to navigate quickly.

Alternate Scenarios:

- The mini-map fails to load or does not align with the property's layout.

Postcondition: The user successfully navigates the property using wayfinding assistance.

2.2.8 3D Floor Structuring

Actors: System

Precondition: Image data must be processed, and the system must generate a basic 3D structure.

Description: The system creates a detailed 3D floor plan using architectural data, allowing for accurate room placement and measurements.

Main Success Scenario:

- The system retrieves architectural data.
- The system generates an interactive 3D floor plan with real-time measurements.
- Users can view and interact with the floor plan in VR.

Postcondition: A 3D floor structure of the property is generated and available for navigation.

2.2.9 Database Interaction and Data Management

Actors: System, Admin

Precondition: The database must be configured (Firebase in this case).

Description: The system handles all interactions with the database, including the upload of images, 3D models, and user login information.

Main Success Scenario:

- The system retrieves data seamlessly from Firebase for 3D model generation.
- Data is stored securely, and version control ensures consistency.

Postcondition: The database is updated with new data and handles requests efficiently.

2.2.10 Optimization of VR Experience (VR Optimization Module)

Actors: System

Precondition: A 3D model must be available in Unity.

Description: The system dynamically loads assets based on user movement to optimize performance, ensuring a smooth experience.

Main Success Scenario:

- Assets are loaded in the background as users explore the property.
- Loading delays are minimized by adjusting the level of detail based on proximity.

Postcondition: Users experience seamless, lag-free navigation.

2.2.11 Property Information Query

Actors: Home buyers, renters, real estate agents

Precondition: The user must be interacting with a virtual property.

Description: Users can query detailed property information, such as pricing, area size, and specifications, while navigating the VR environment.

Main Success Scenario:

- Users click on hotspots or interact with objects in the VR space to view information.

Postcondition: The user successfully views property details in VR.

2.3 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depend on what kind of technique has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes.

2.3.1 Module 1: VR UI Module

Following are the requirements for the VR UI Module:

1. The system should allow users to interact with objects within the VR environment by rotating and zooming.
2. The UI should adapt based on the user's proximity to the object in the virtual space.

2.3.2 Module 2: 3D Environment Modelling

Following are the requirements for the 3D Environment Modelling [1]:

1. The system should create realistic 3D models of rooms and external surroundings, integrating weather conditions and day/night cycles.
2. Lighting and sound effects should adjust dynamically based on the time of day and user actions within the virtual space.

2.4 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable. The following are the non-functional requirements for the platform.

2.4.1 Reliability

The platform must ensure minimal downtime, targeting an availability rate of 99.9

2.4.2 Usability

The platform should be easy to use, with intuitive controls for navigation and interaction within the VR environment. It should also cater to users unfamiliar with VR technology, ensuring ease of learning, error avoidance, and recovery. The user interface should be designed to create an optimum user experience.

2.4.3 Performance

The VR system should load assets quickly, with minimal delays. Loading times for property environments should not exceed 2 seconds to ensure a seamless user experience.

2.4.4 Security

User data, including personal information and virtual tour activity, must be protected using encryption protocols to prevent unauthorized access and data breaches.

2.5 Domain Model

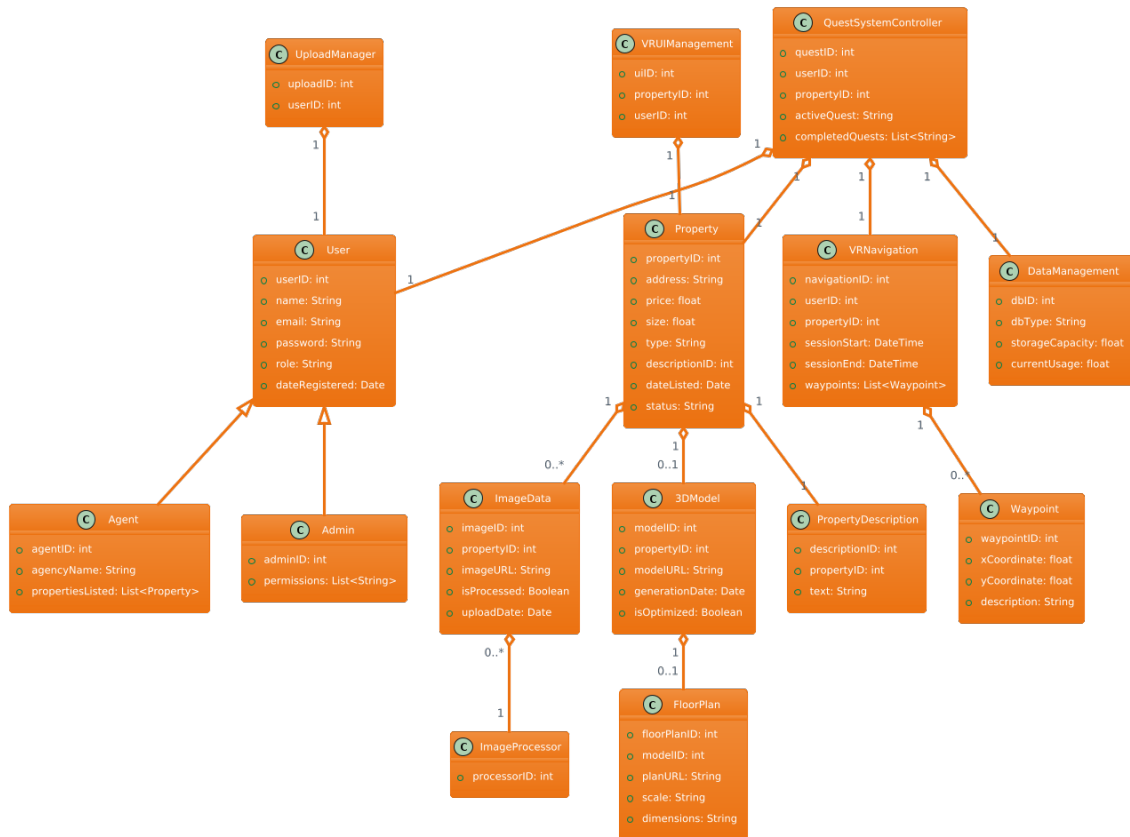


Figure 2.2: Domain Model.

Chapter 3

System Overview

3.1 Architectural Design

The system follows a layered architecture, where the application is divided into distinct layers for better scalability, maintainability, and separation of concerns. The presentation layer handles the rendering of the VR environment, allowing users to navigate through properties. This layer communicates with the business logic layer, which processes user interactions, manages navigation, and retrieves property data. The data layer is responsible for storing and managing 3D models and property details, which are hosted on Firebase for scalable and secure cloud-based storage. The layered design ensures modularity and dynamic retrieval of necessary data during user navigation.

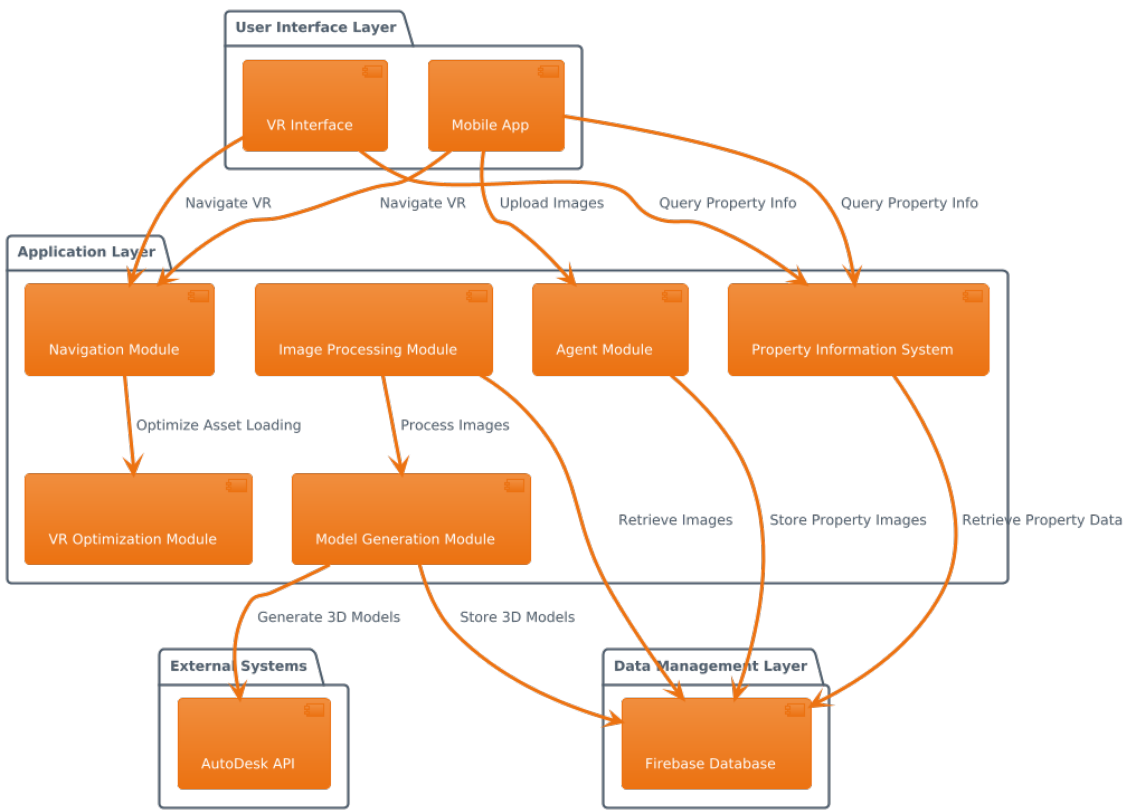


Figure 3.1: Architecture Diagram 1.

3.2 Design Models

Design Models for Object Oriented Development Approach

3.2.1 Activity Diagrams

- Activity Diagram 1

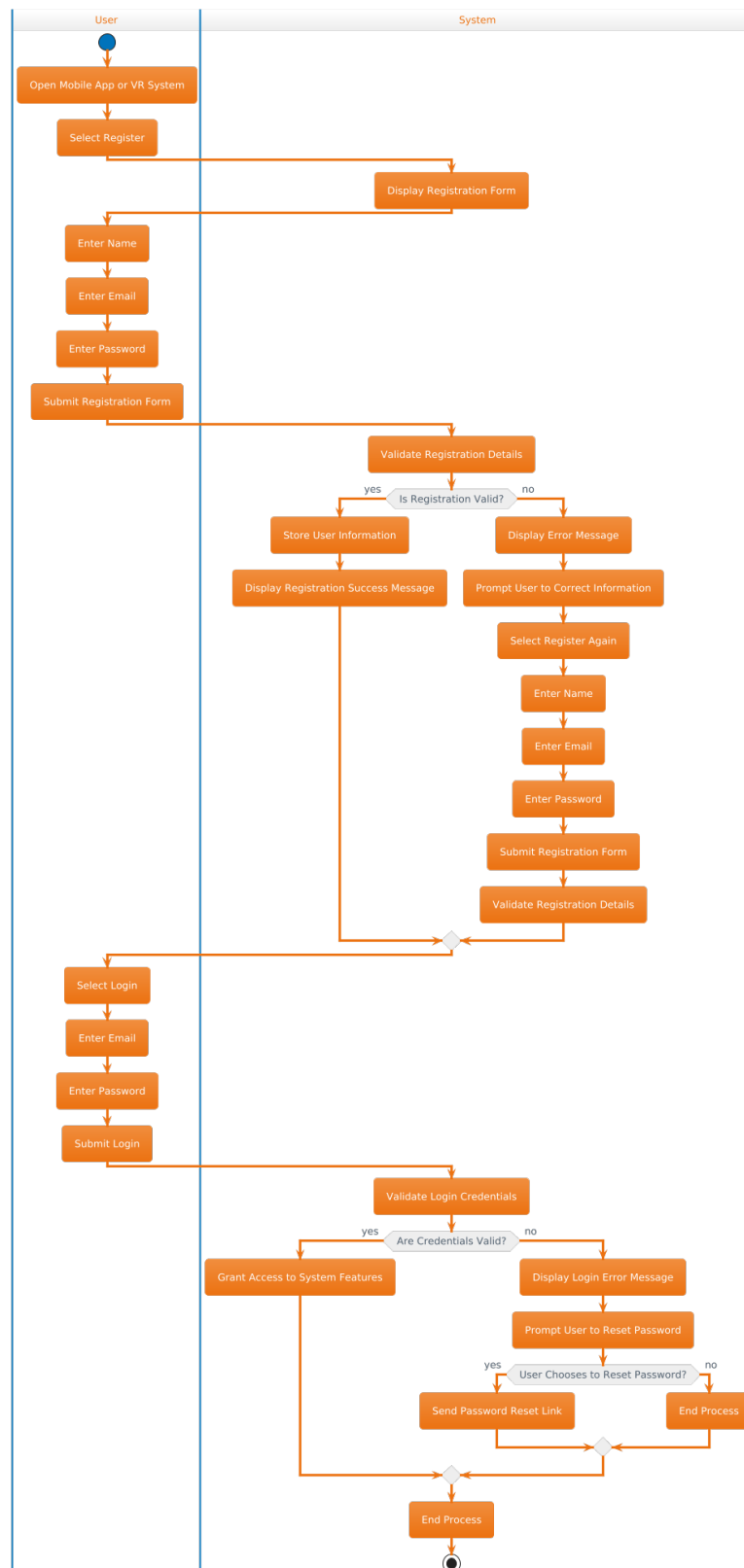


Figure 3.2: Activity Diagram 1.

- Activity Diagram 2

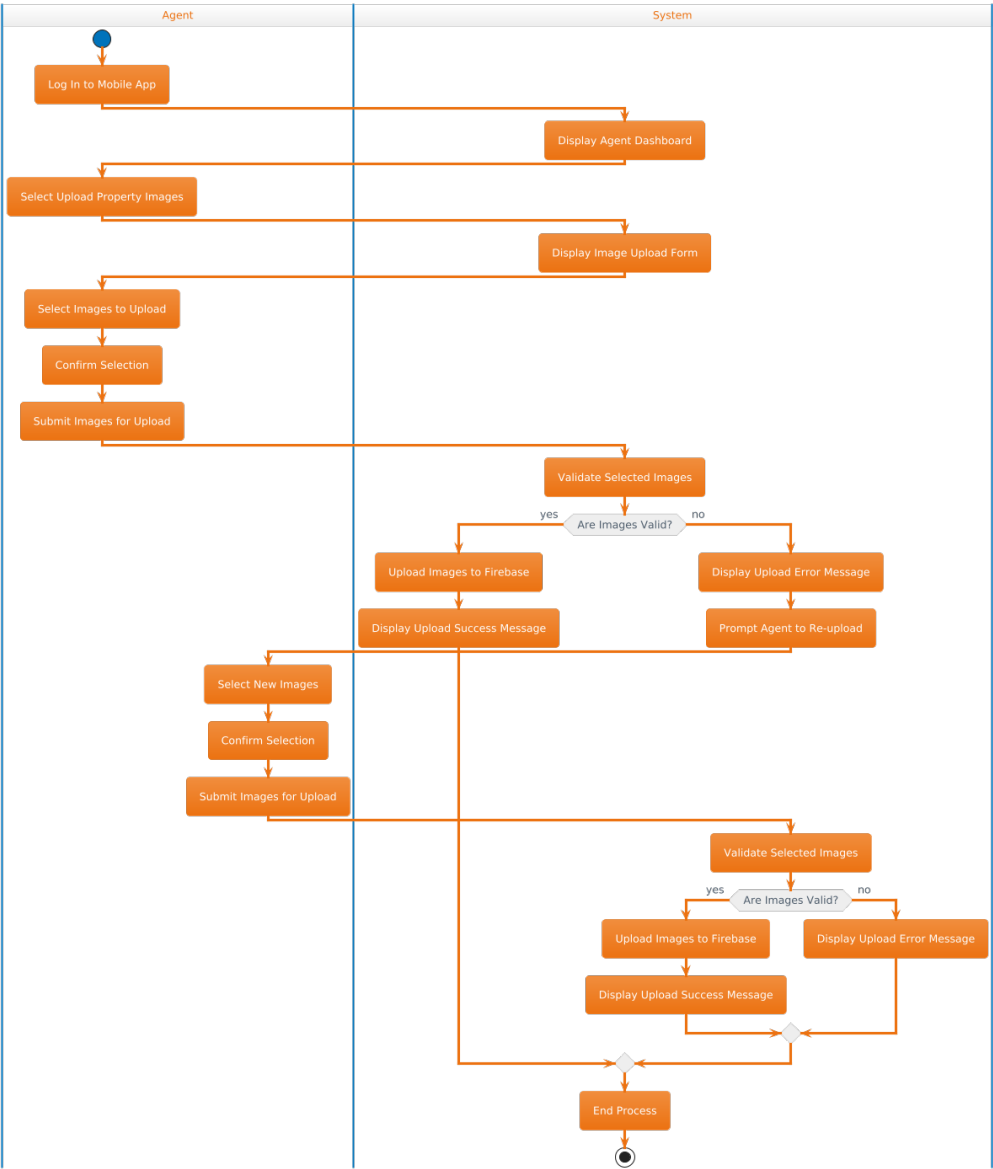


Figure 3.3: Activity Diagram 2.

- Activity Diagram 3

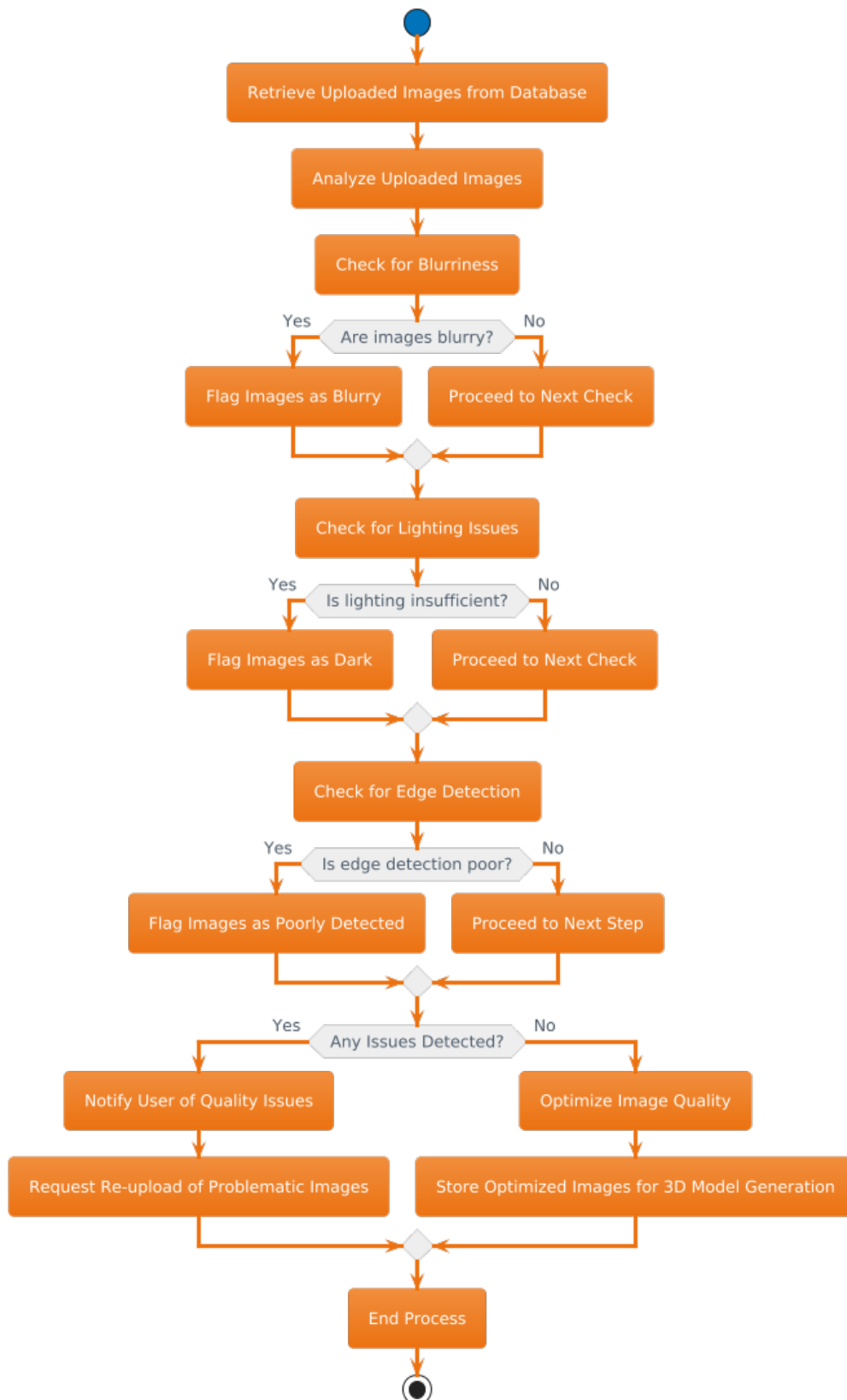


Figure 3.4: Activity Diagram 3.

• Activity Diagram 4

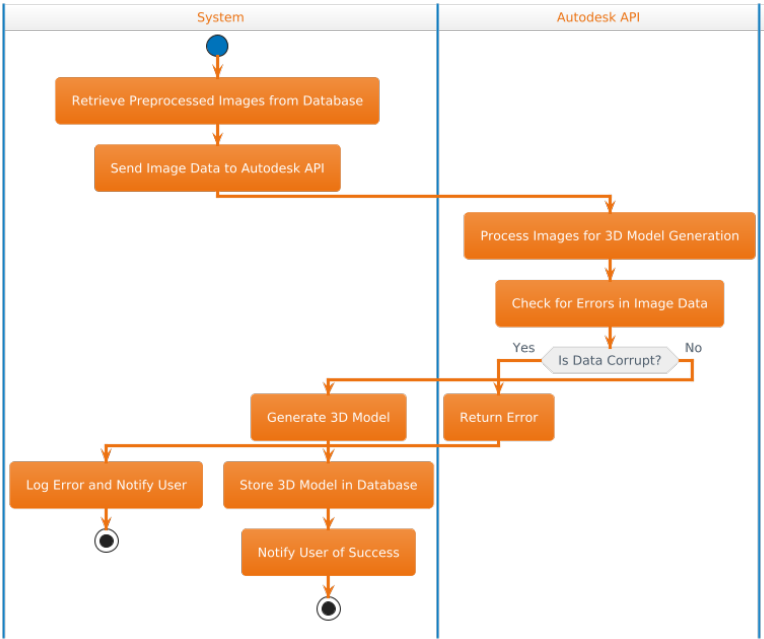


Figure 3.5: Activity Diagram 4.

• Activity Diagram 5

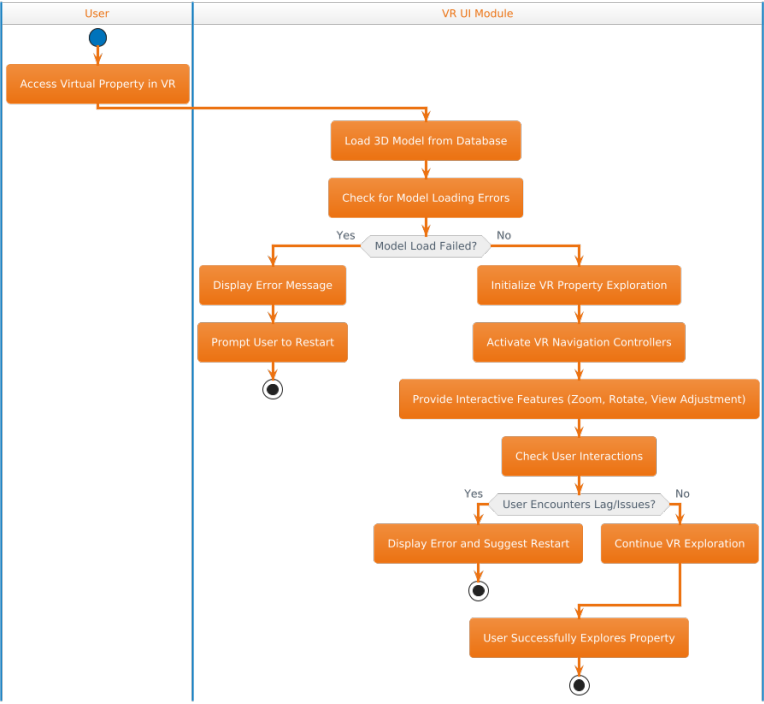


Figure 3.6: Activity Diagram 5.

- Activity Diagram 6

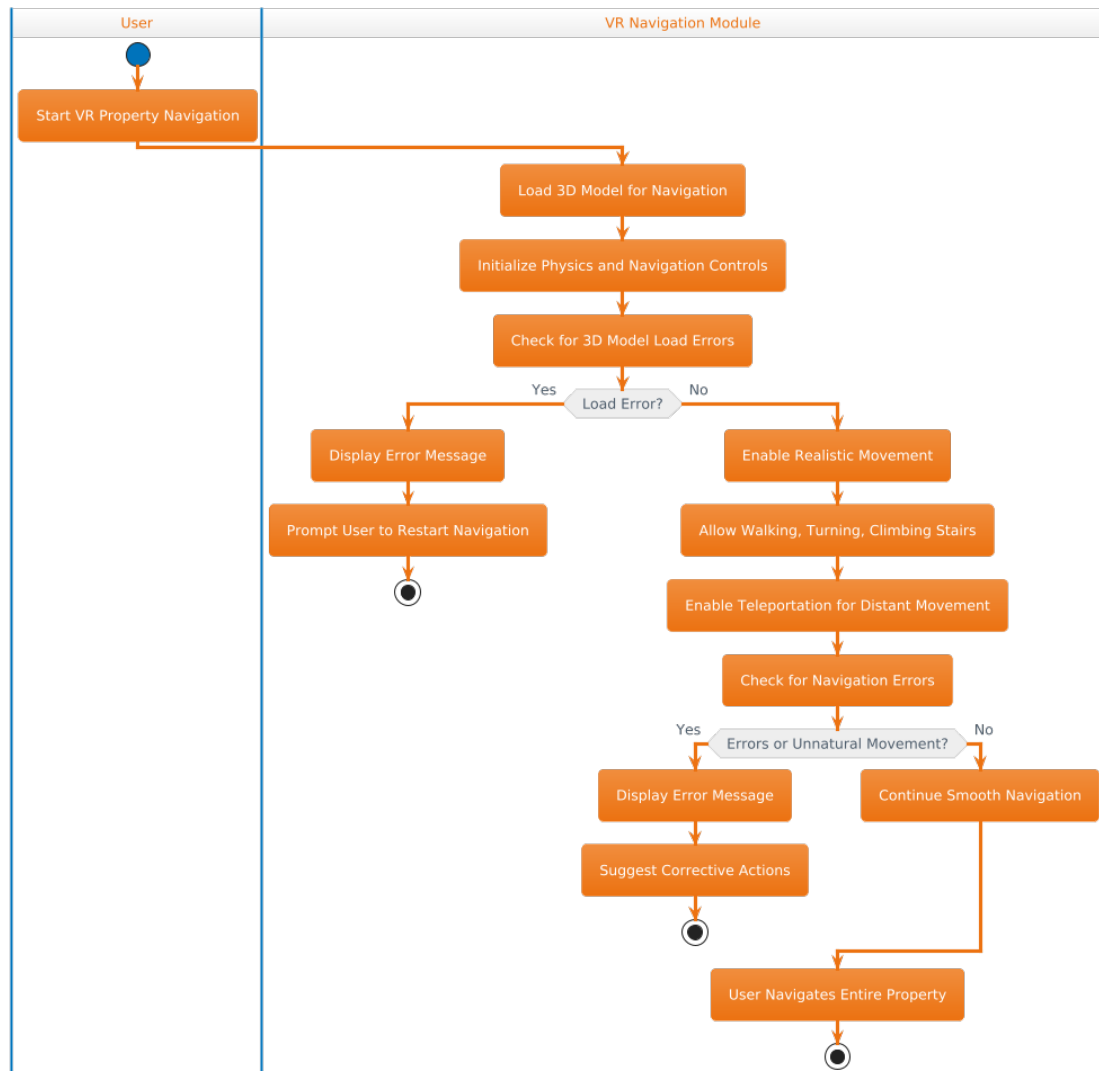


Figure 3.7: Activity Diagram 6.

- Activity Diagram 7

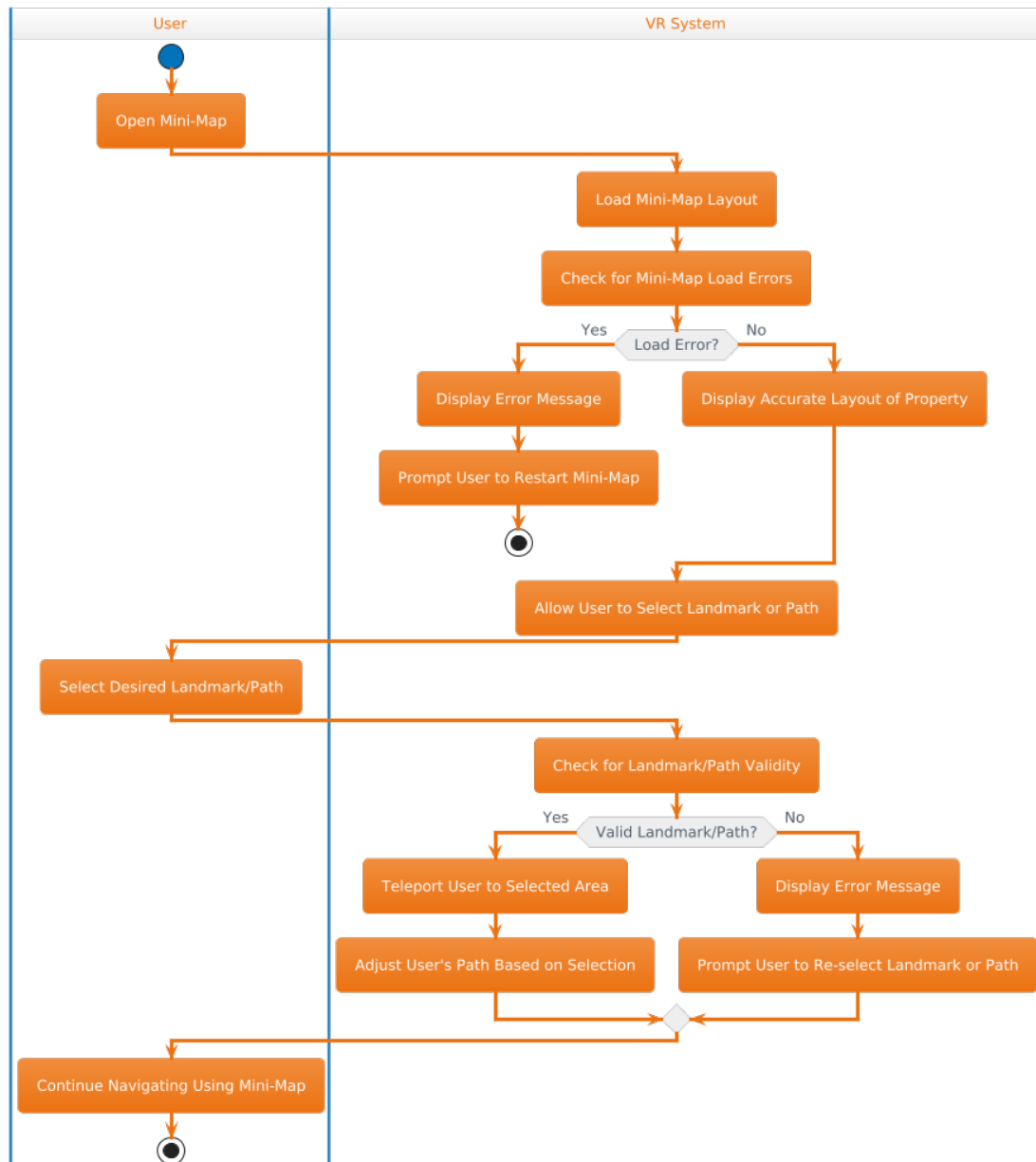


Figure 3.8: Activity Diagram 7.

- Activity Diagram 8

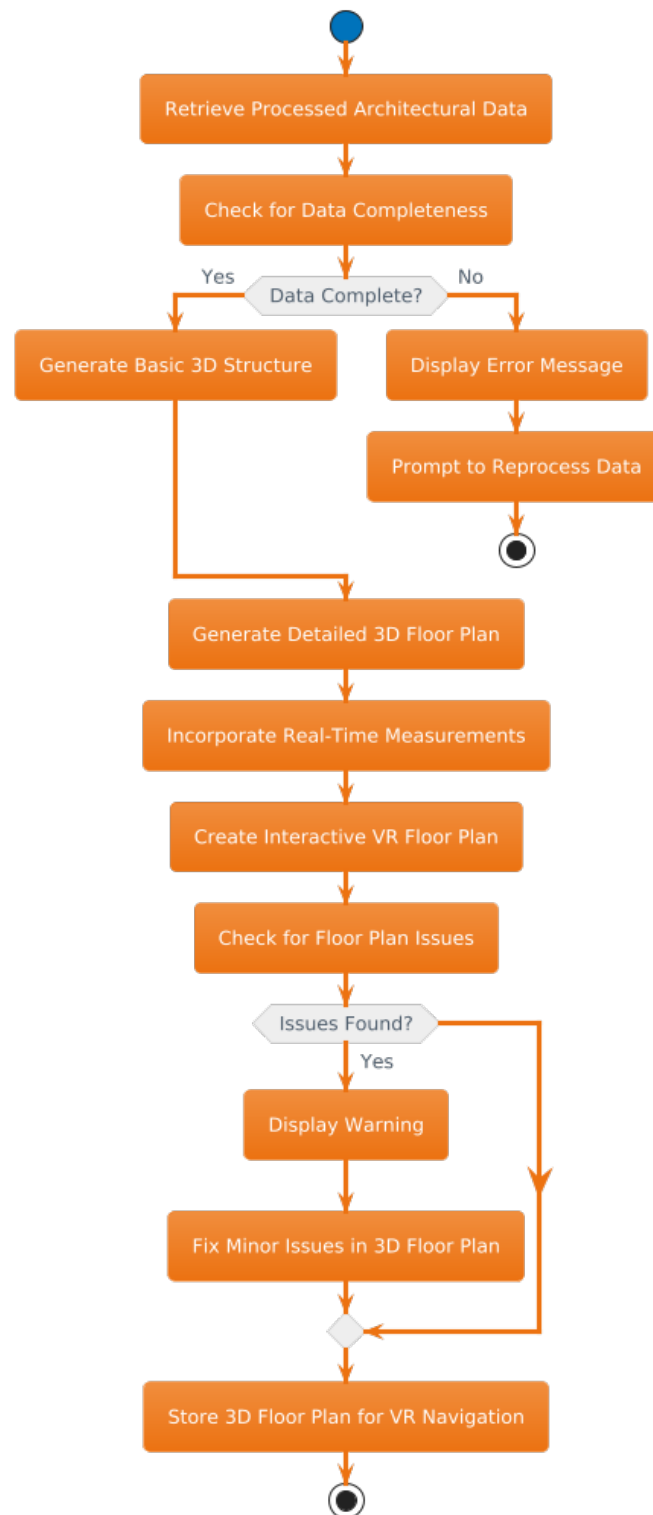


Figure 3.9: Activity Diagram 8.

- Activity Diagram 9

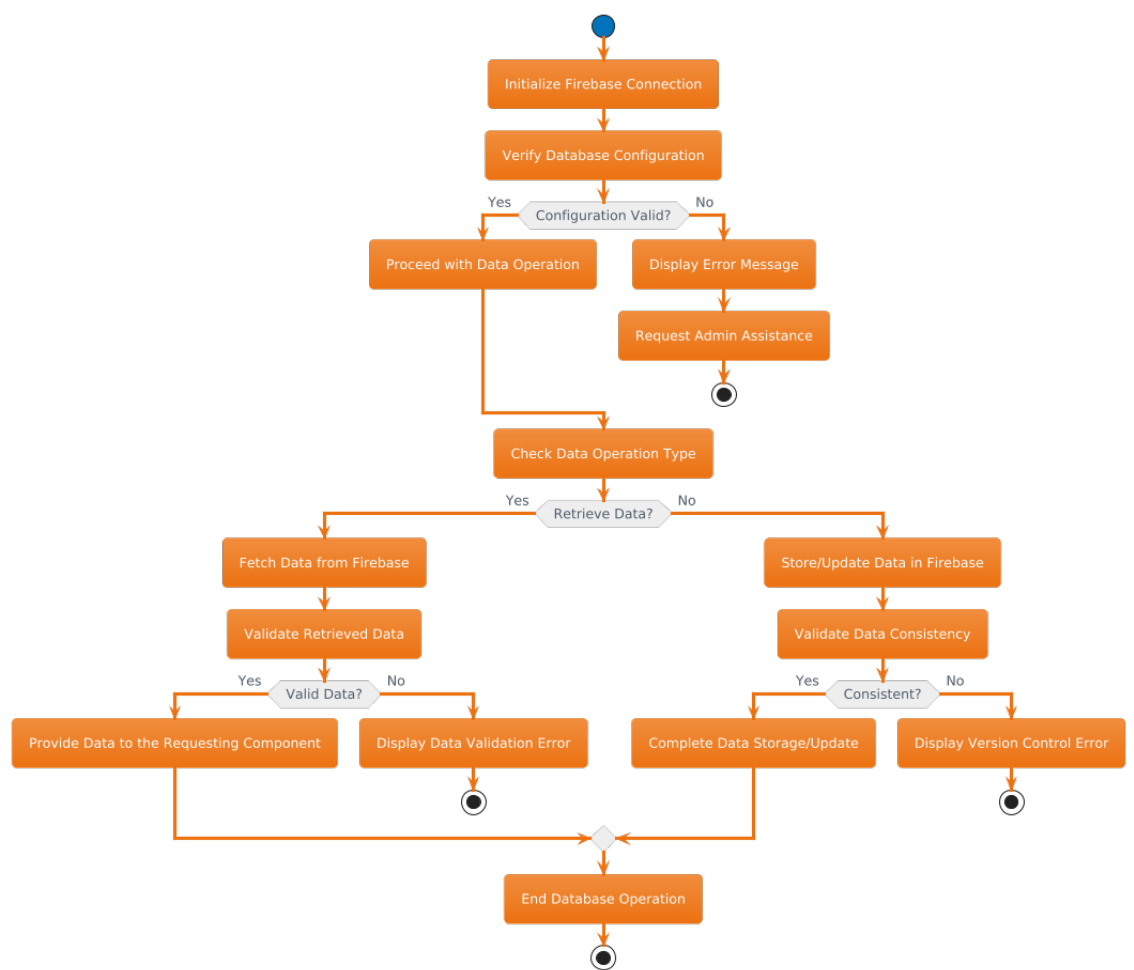


Figure 3.10: Activity Diagram 9.

- Activity Diagram 10

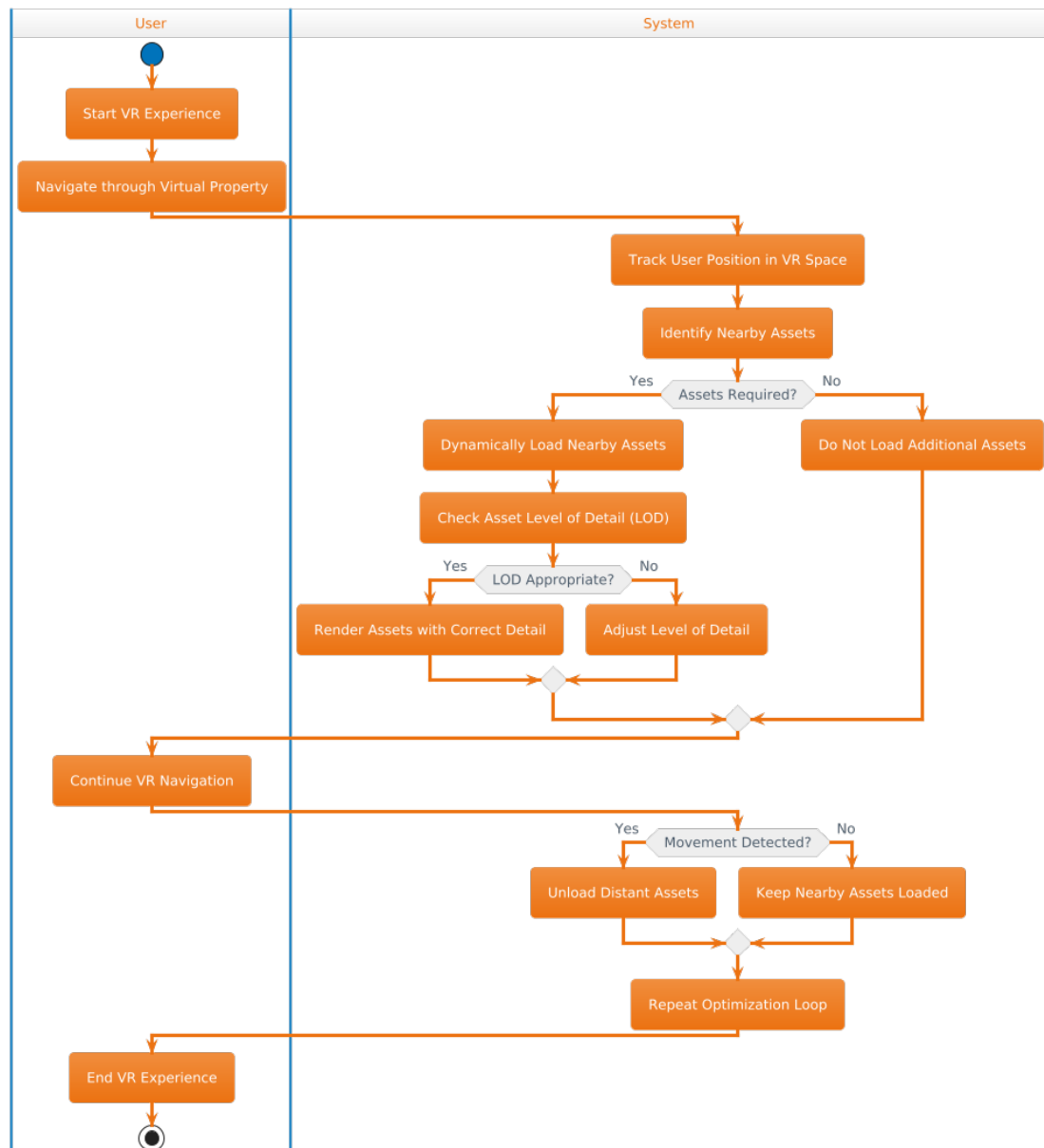


Figure 3.11: Activity Diagram 10.

- Activity Diagram 11

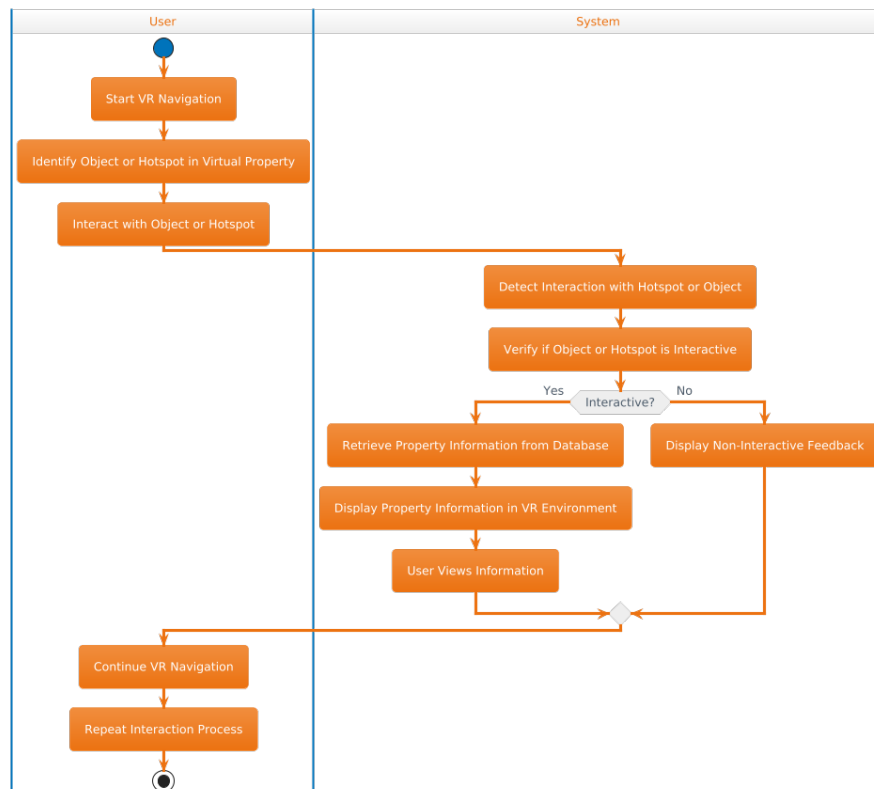


Figure 3.12: Activity Diagram 11.

3.2.2 Class Diagram

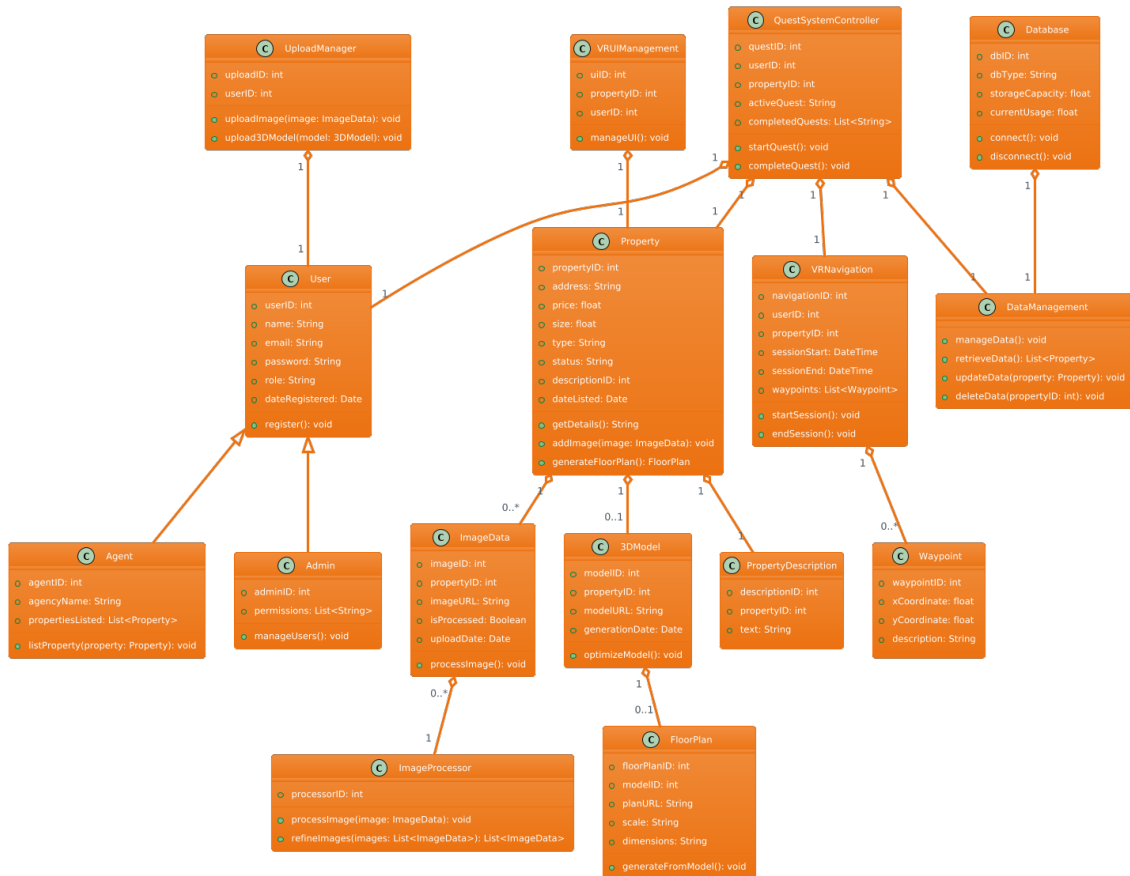


Figure 3.13: Class Diagram.

3.2.3 Class Level Sequence Diagrams

- Sequence Diagram 1

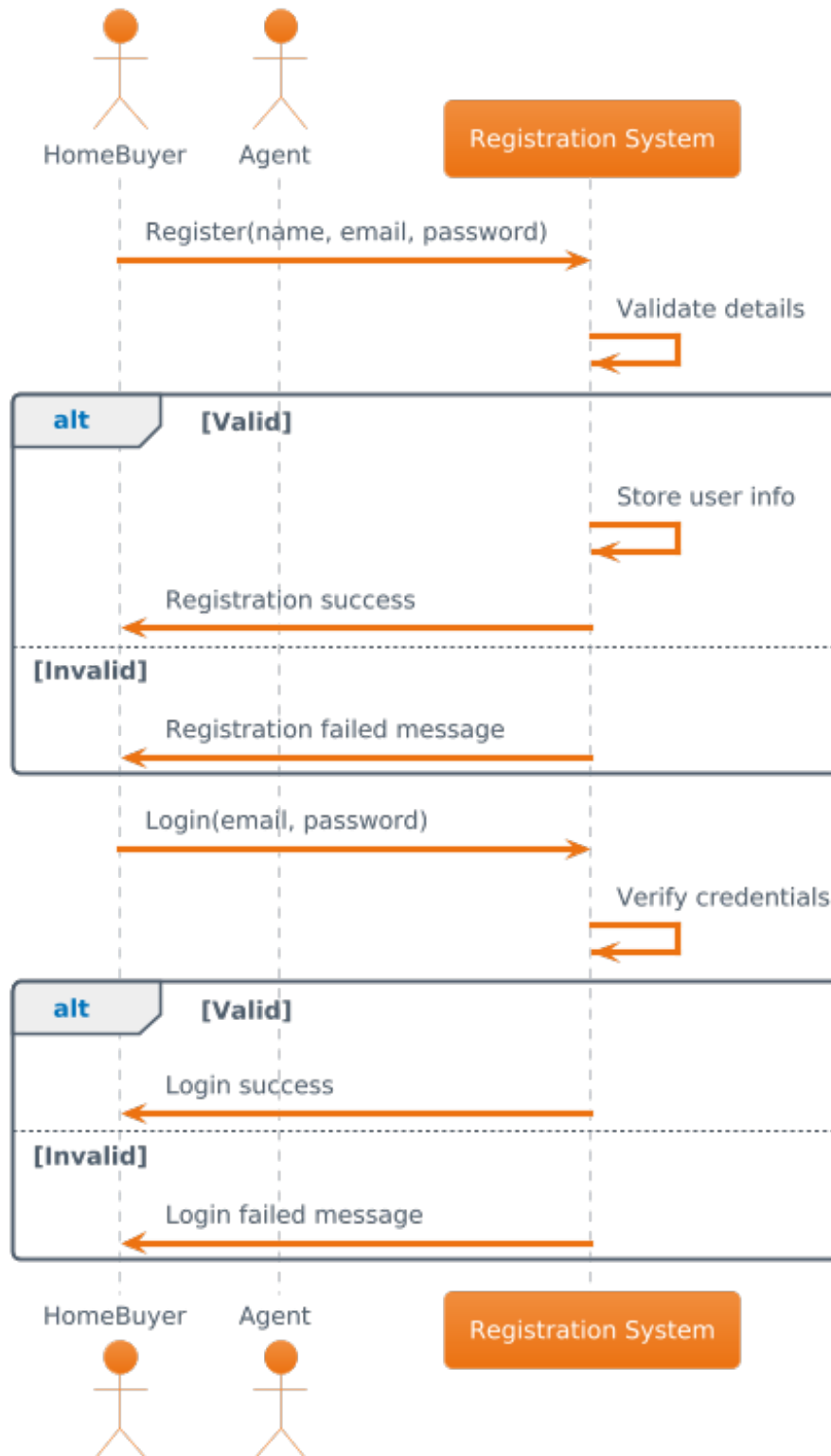


Figure 3.14: Sequence Diagram 1.

- Sequence Diagram 2

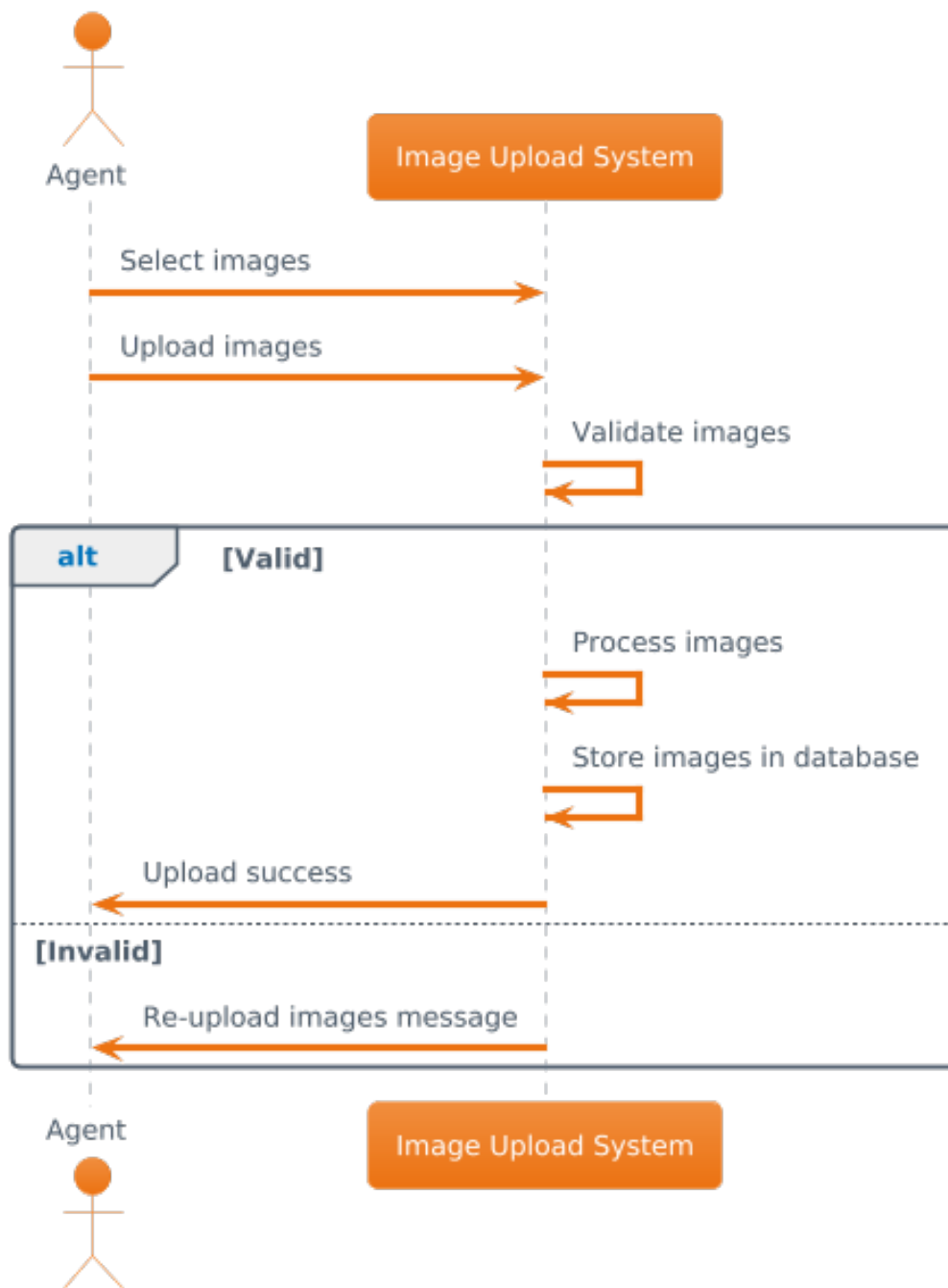


Figure 3.15: Sequence Diagram 2.

- Sequence Diagram 3

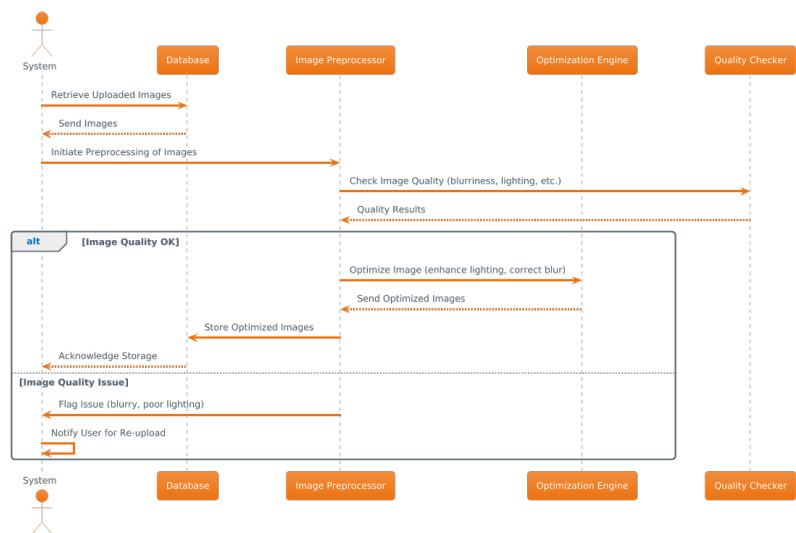


Figure 3.16: Sequence Diagram 3.

• Sequence Diagram 4

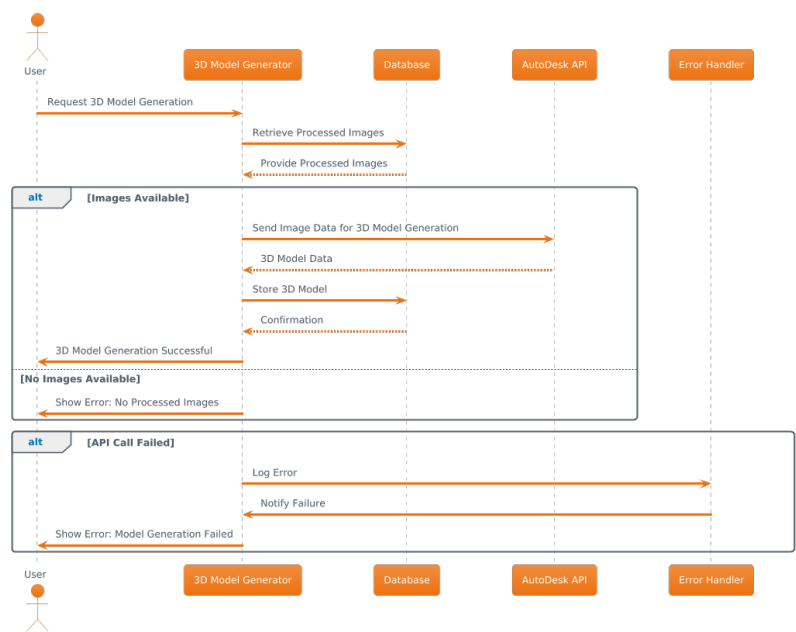


Figure 3.17: Sequence Diagram 4.

• Sequence Diagram 5

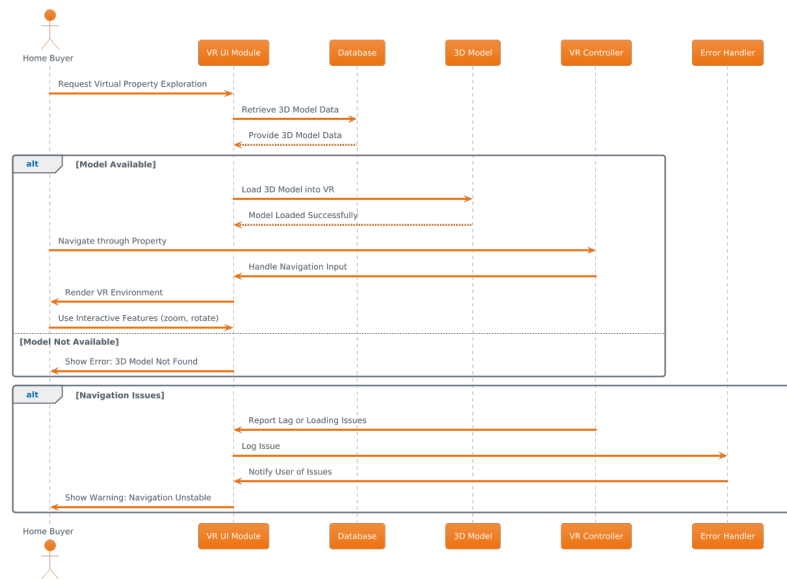


Figure 3.18: Sequence Diagram 5.

- Sequence Diagram 6

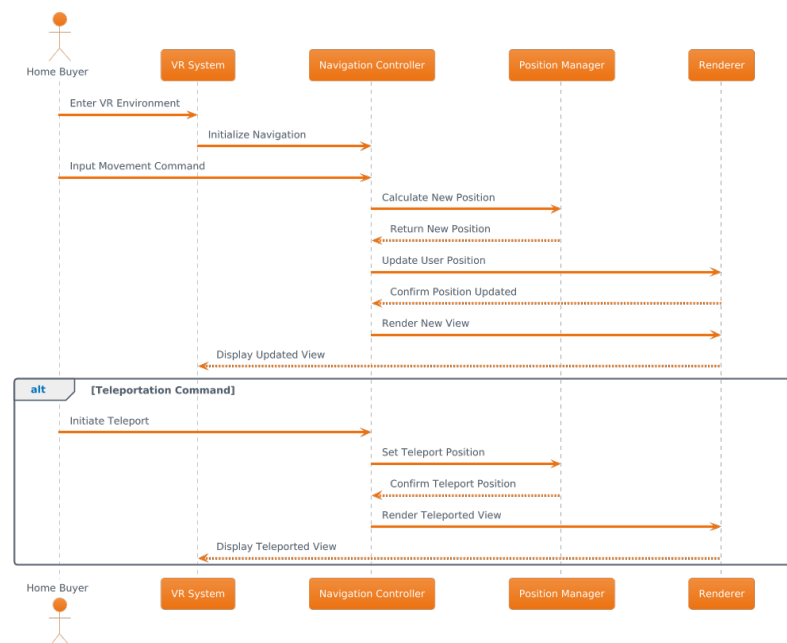


Figure 3.19: Sequence Diagram 6.

- Sequence Diagram 7

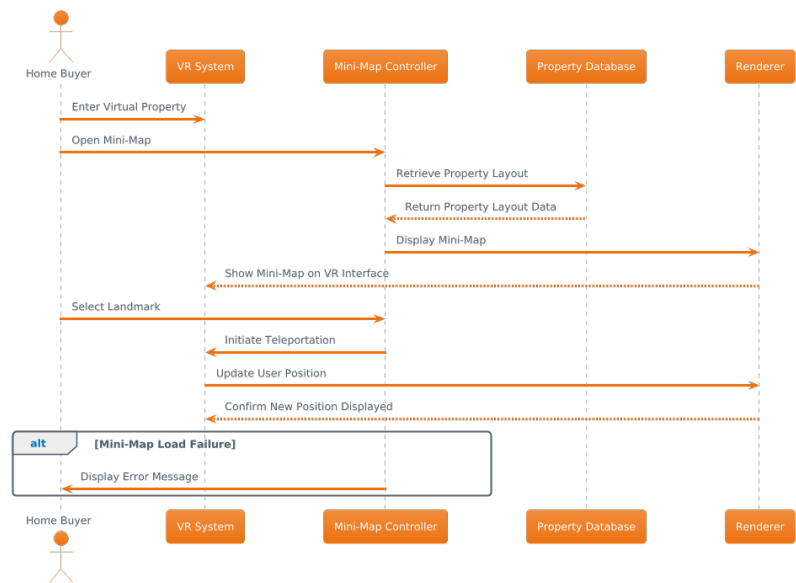


Figure 3.20: Sequence Diagram 7.

• Sequence Diagram 8

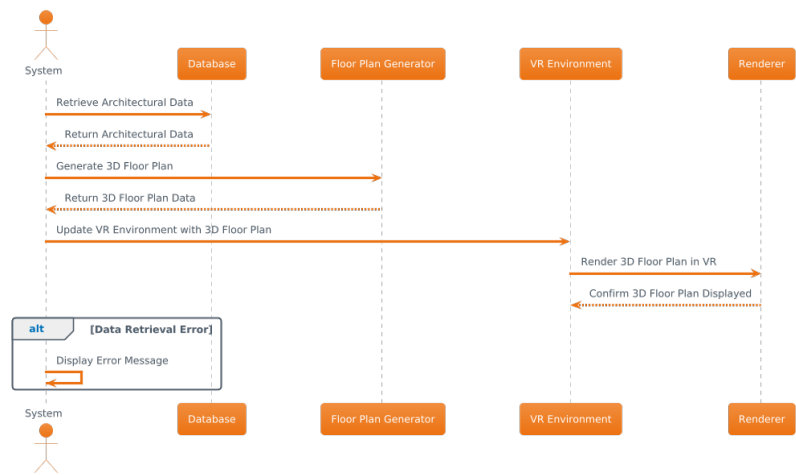


Figure 3.21: Sequence Diagram 8.

• Sequence Diagram 9

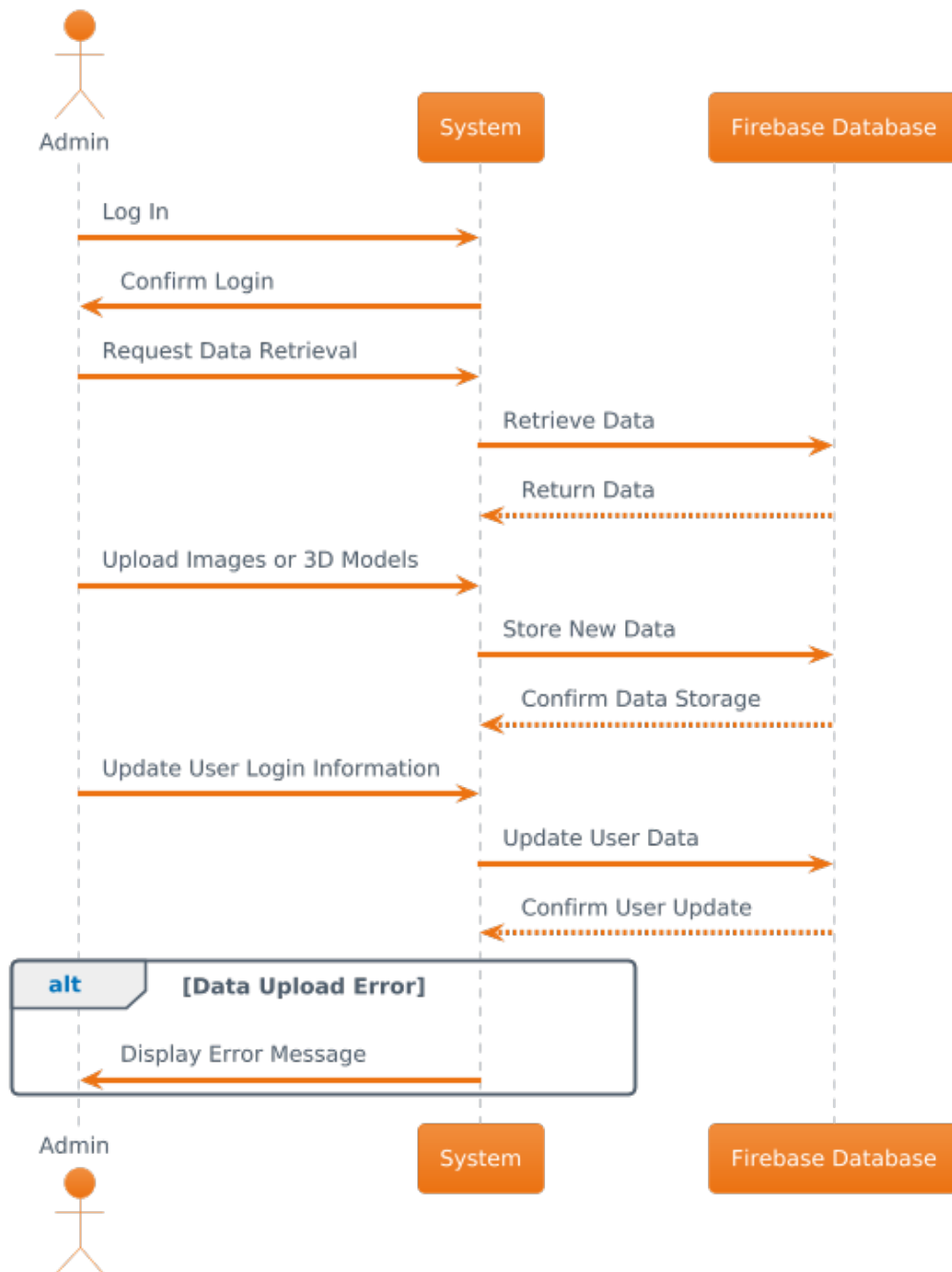


Figure 3.22: Sequence Diagram 9.

- Sequence Diagram 10

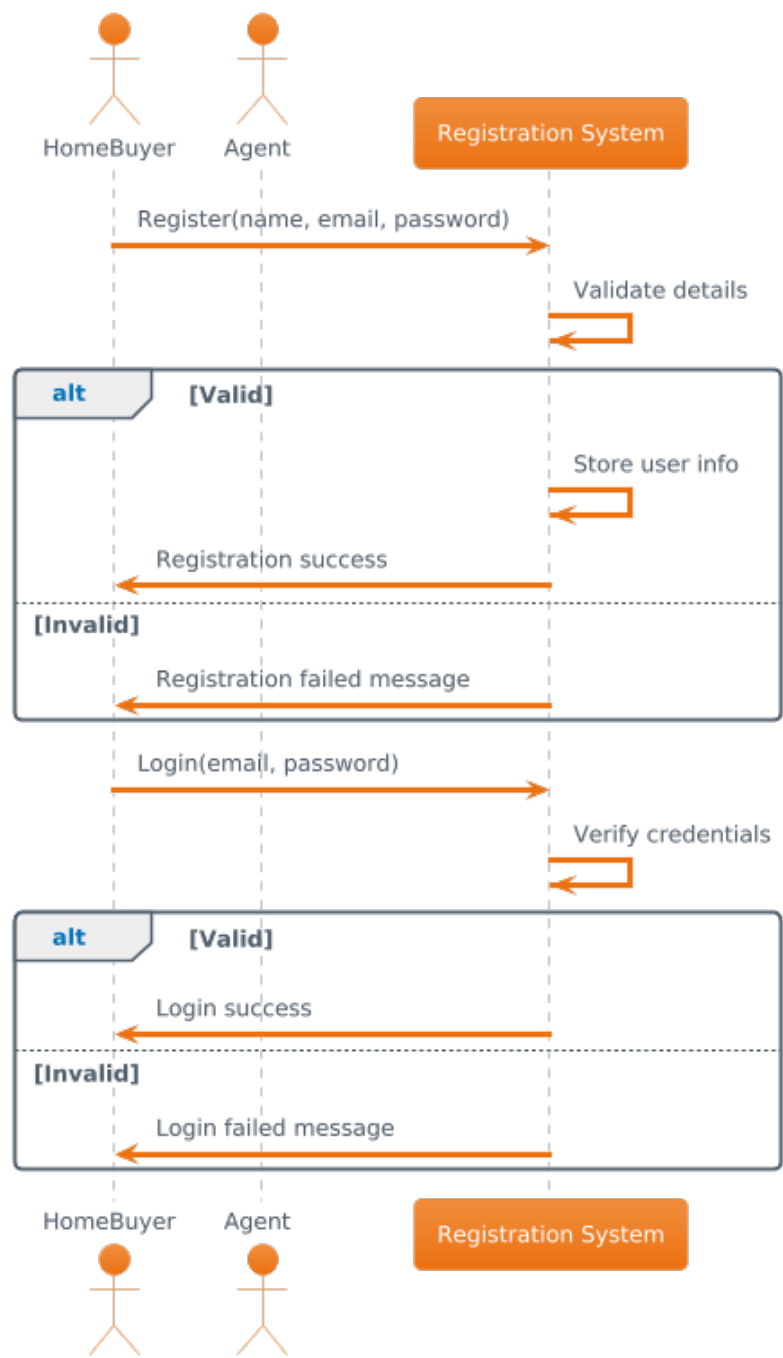


Figure 3.23: Sequence Diagram 1.

- Sequence Diagram 11

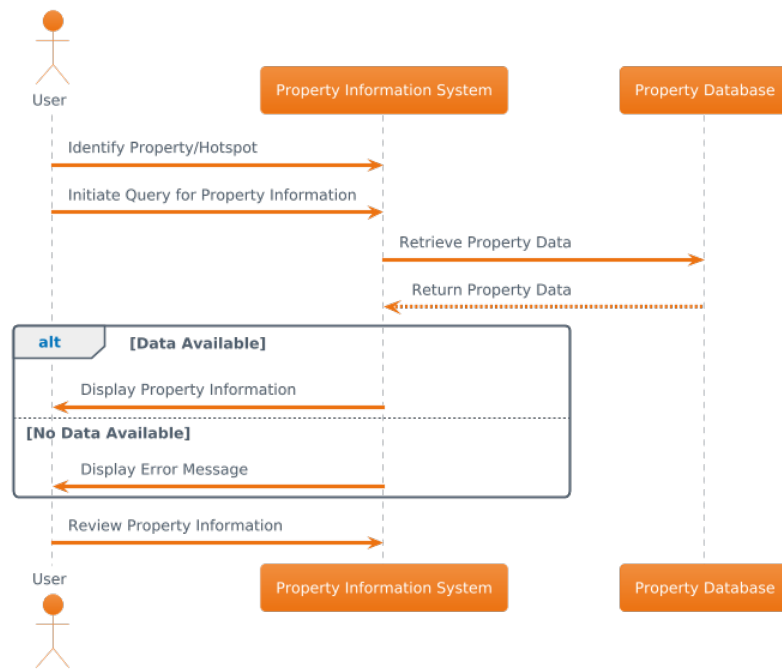


Figure 3.24: Sequence Diagram 11.

3.2.4 State Transition Diagram

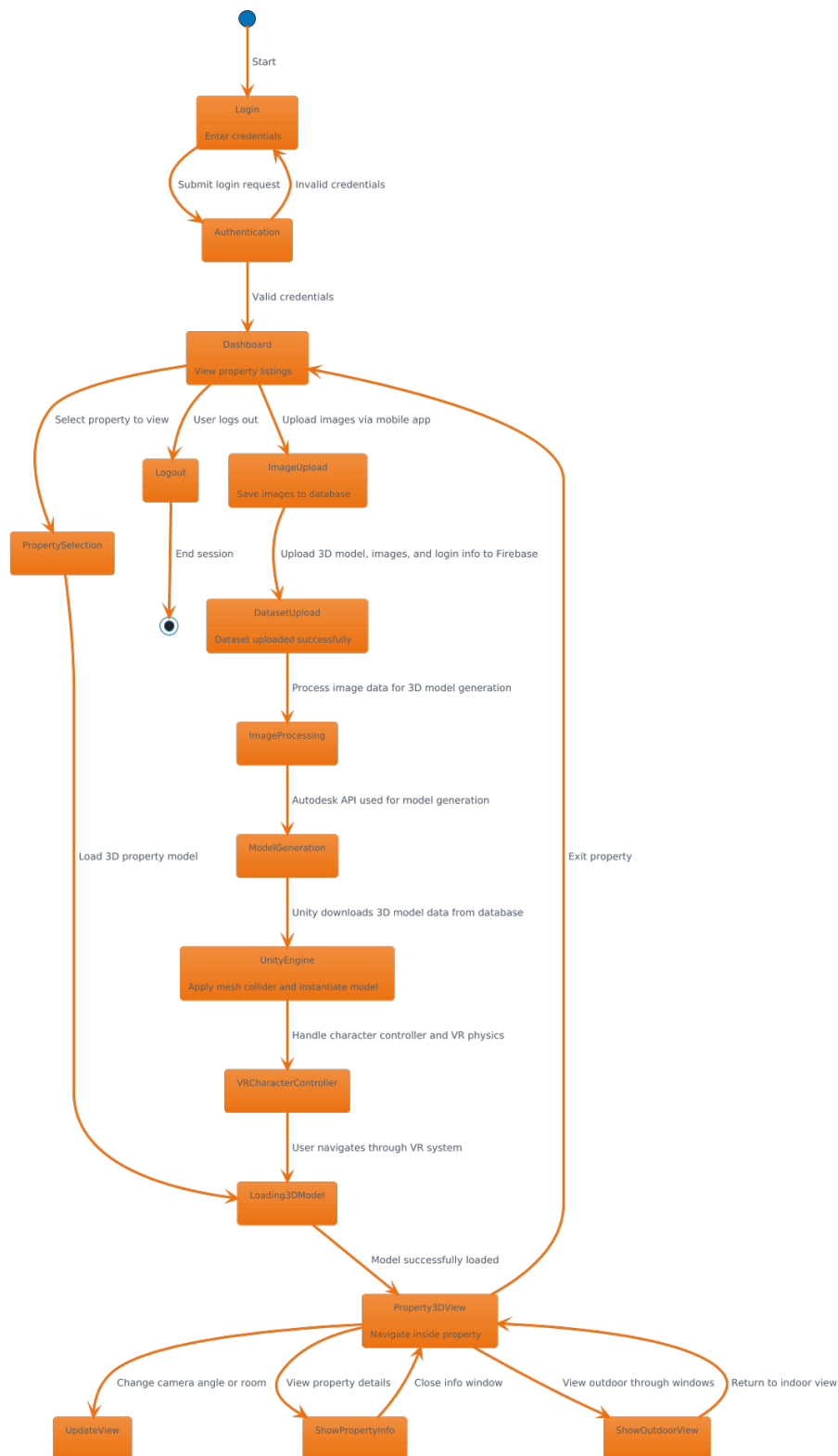


Figure 3.25: State Transition Diagram.

3.3 Data Design

Data within Palantir360 includes property models, user profiles, and session information. All data is stored in a cloud-based Firebase database, ensuring scalability. The data design prioritizes efficient retrieval and storage of 3D models, with indexing for quick access during user navigation. Real-time updates, such as property details or environmental changes, are reflected in the VR environment using asynchronous data handling

3.3.1 Entity Relationship Diagram

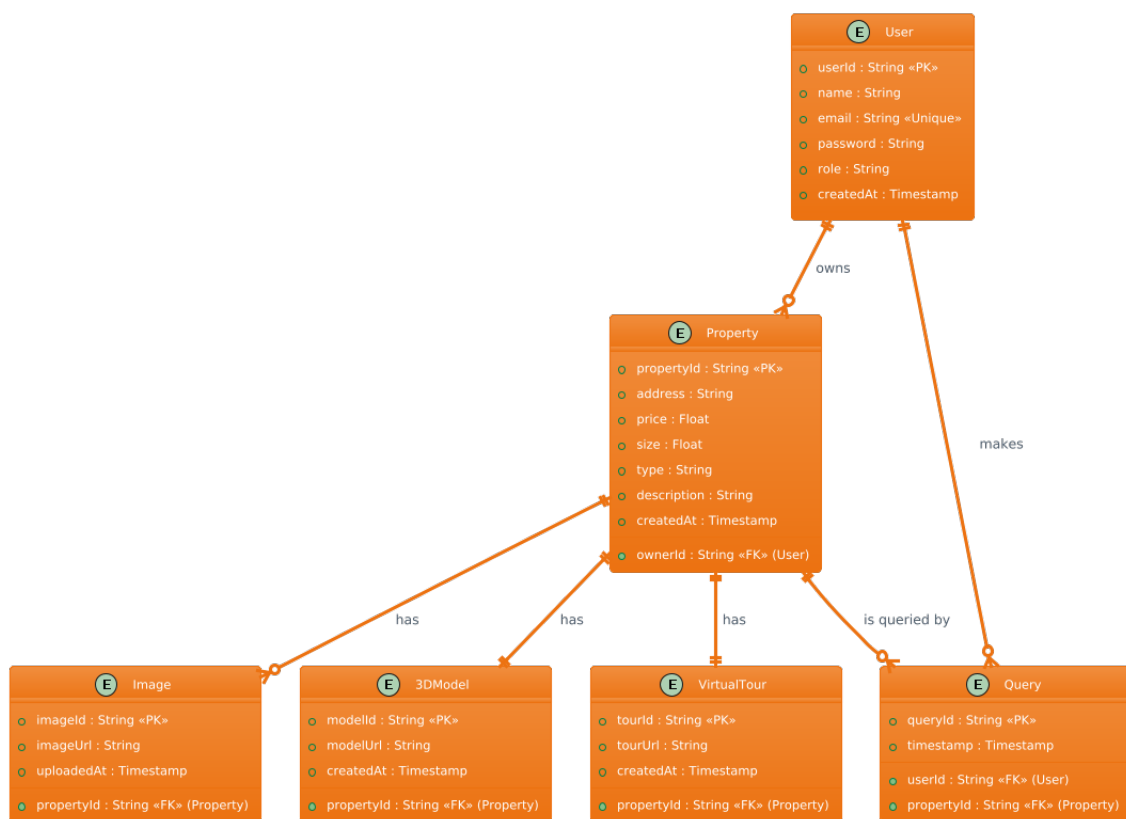


Figure 3.26: Entity Relationship Diagram.

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- [1] INRIA FunGraph Team. 3d gaussian splatting for real-time rendering. <https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/>, 2024. Accessed: 2024-10-06.
- [2] NeRF Studio Team. Nerf studio documentation. <https://docs.nerf.studio/>, 2024. Accessed: 2024-10-06.